LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	ANKUSH GUPTA
Roll Number	MRA2026003
Programme / Department	MTech in IT with specialization in Robotics and AI
Topic ID	T7_
Full Assigned Topic	Knowledge Cutoff Effects on LLM-Generated Literature Reviews in Rapidly Evolving Fields
AI Tool Used	Claude
Date	21/08/2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the "best" paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations
B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ■ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☐ Once or twice ■ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☐ Rarely ■ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐ 3 ■ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Claude
Model name	Sonnet 5
Model/version shown	NOT SHOWN
Access tier	■ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	■ Yes ☐ No ☐ Not sure
Was web browsing actually used?	■ Yes ☐ No ☐ Not sure

Research/Deep Research mode used?	☐ Yes ■ No
Date/time generated	9:45 21/08/2026

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

> Generate a scholarly research paper on the topic: "[YOUR ASSIGNED TOPIC]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

■ Save the complete AI-generated paper

■ Save the complete reference list

■ Take screenshot(s) showing the generated output/references

■ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Knowledge Cutoff Effects on LLM-Generated Literature Reviews in Rapidly Evolving Fields.

Measure	Value
Approximate pages	10
Approximate word count	4175
Number of sections	7
Total number of references	19
Approximate in-text citation occurrences	34
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ■ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	19
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	11
URL only	0

No persistent identifier supplied	0
TOTAL	30

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	2402.01788
R2	2412.15249
R3	2407.18940
R4	2508.06401
R5	2312.10997
R6	2305.15186
R7	2502.14776
R8	NIL
R9	NIL
R10	NIL

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Completely Convincing	5	Yes
R2	Completely Convincing	5	Yes
R3	Completely Convincing	5	Yes
R4	Looks Genuine	4	Yes
R5	Completely Convincing	5	Yes
R6	Looks Genuine	4	Yes
R7	Completely Convincing	5	Yes
R8	Completely Convincing	5	Yes
R9	Completely Convincing	5	Yes
R10	Completely Convincing	5	Yes

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

☐ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	N	Y
R2	Y	N	Y	Y	N	Y
R3	Y	Y	Y	Y	Y	Y
R4	Y	Y	Y	Y	N	Y
R5	Y	Y	Y	Y	N	Y
R6	Y	Y	Y	Y	N	Y
R7	Y	Y	Y	Y	N	Y
R8	Y	Y	Y	Y	Y	NA
R9	Y	Y	Y	Y	N	NA
R10	Y	Y	Y	Y	Y	NA

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	https://arxiv.org/	Yes	Publication exists and important metadata are correct.
R2	https://arxiv.org/	Yes	Publication exists and important metadata are correct.
R3	https://arxiv.org/	Yes	Publication exists and important metadata are correct.
R4	https://arxiv.org/	Yes	Publication exists and important metadata are correct.
R5	https://arxiv.org/	Yes	Publication exists and important metadata are correct.
R6	https://arxiv.org/	Yes	Publication exists and important metadata are correct.
R7	https://arxiv.org/	Yes	Publication exists and important metadata are correct.

R8	https://dl.acm.org/	Not Applicable	Publication exists, but one or more bibliographic details are incorrect.
R9	https://academic.oup.com/	Not Applicable	Publication exists, but one or more bibliographic details are incorrect.
R10	https://aclanthology.org/	Not Applicable	Publication exists, but one or more bibliographic details are incorrect.

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	Publication exists and important metadata are correct.
R2	A	Publication exists and important metadata are correct.
R3	A	Publication exists and important metadata are correct.
R4	A	Publication exists and important metadata are correct.
R5	A	Publication exists and important metadata are correct.
R6	A	Publication exists and important metadata are correct.
R7	A	Publication exists and important metadata are correct.
R8	B	Publication exists, but one or more bibliographic details are incorrect.
R9	B	Publication exists, but one or more bibliographic details are incorrect.
R10	B	PPublication exists, but one or more bibliographic details are incorrect.

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 8 B =2 C =0 D = 0 E =0

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = 4A + 3B + C + E

Authenticity Score = [(4A + 3B + C + E) / (4N)] x 100

My calculation: 4(8) + 3(2) + 1(0) + 0(0) + 1(1) = 38 points

Authenticity Score = 38 / (4 x 10) x 100 = 95/100

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk

Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = 10   Total predictions = 10   Prediction Accuracy = 100%

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	19
References deeply audited	10
A: Verified	8
B: Wrong metadata	2
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	0
Authenticity Score	_95 /100
Prediction Accuracy	100_ %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

In this audit, the professional-looking citations were genuine,but venue information was not mentioned and some of the citations are not published yet.

2. Did all genuine references actually support the claims for which they were cited?

References 3, 8 and 10 supported the claims made for them. Remaining References supported the general evaluation claim, but names of authors are given in short forms and in some of the cases venue is not mentioned as well as identification numbers are also missing in the last 3 references.

3. What was the most serious citation-integrity problem you observed?

In last 3 references identification numbers are missing, in this scenario it could be difficult to trust the authenticity of the citation.

4. What is the single most important lesson you learned from this lab?

A citation could be genuine and still be used incorrectly. Research auditing must verify both that the source exists and that it actually supports the claim.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: Ankush Gupta

Roll Number: MRA2026003

Signature: 

Date: 21/08/2026

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support Verification -> Quantitative Scores -> Submission

Knowledge cutoff effects on LLM-generated literature reviews

Generate a scholarly research paper on the topic: "Knowledge Cutoff Effects on LLM-Generated Literature Reviews in Rapidly Evolving Fields". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

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The paper is a 9-page Word document covering all requested sections — Title, Abstract, Keywords, Introduction, Background/Related Work, Main Technical Discussion (mechanisms of cutoff-driven distortion), Current Approaches (RAG, LitLLM, ChatCite, OpenScholar, quality-control pipelines), Research Challenges/Limitations, Research Gaps/Future Directions, Conclusion, and a References list of 18 genuine, verifiable sources (arXiv IDs, DOIs, PMID, and venue details included) with in-text citations throughout.

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Knowledge Cutoff Effects on LLM-Generated Literature Reviews in Rapidly Evolving Fields

A Review of Mechanisms, Current Mitigations, and Open Research Problems

Abstract

Large language models (LLMs) are increasingly used to draft, summarize, and even automate scientific literature reviews, yet every such model carries a fixed knowledge cutoff beyond which it has no direct exposure to newly published work. This paper examines how knowledge cutoff effects distort LLM-generated literature reviews specifically within rapidly evolving research fields, where publication velocity, terminological drift, and citation lag compress the effective shelf life of a model's parametric knowledge. Drawing on recent empirical work on effective versus reported cutoffs, temporal knowledge decay, citation hallucination, and retrieval-augmented generation (RAG), the paper synthesizes evidence that cutoff effects manifest not as a single sharp boundary but as a graded, resource-dependent degradation that produces recency bias, stale framing of "open problems," omission of superseding work, and fabricated or outdated citations. Current mitigation strategies—retrieval augmentation, hybrid keyword-embedding search, plan-then-write generation pipelines, and post-hoc citation verification—reduce but do not eliminate these effects, particularly for the most recent months of a field's literature, where indexing lag and retrieval sparsity persist. The paper concludes by identifying research gaps around dynamic cutoff auditing, field-adaptive retrieval scheduling, and standardized benchmarks for time-sensitive review generation, and argues that treating knowledge cutoff as a first-class, quantifiable variable is necessary for the responsible use of LLMs in scholarly synthesis.

Keywords: large language models; knowledge cutoff; literature review generation; retrieval-augmented generation; citation hallucination; temporal generalization; scientific writing automation

1. Introduction

Literature reviews are among the most time-consuming yet foundational activities in scholarly research: they establish the boundary of known work, situate a contribution relative to prior art, and identify open problems. As the volume of scientific publication has grown, researchers have increasingly turned to large language models (LLMs) to accelerate literature discovery, summarization, and synthesis (Tang et al., 2025). At the same time, every LLM is trained on a corpus assembled up to a particular point in time — its knowledge cutoff — after which the model has no direct parametric awareness of newly published findings, terminology, or shifts in scientific consensus (Cheng et al., 2024).

This limitation is not merely inconvenient; it is potentially distorting when the object of the task is itself the state of a fast-moving literature. Fields such as large language model research, generative AI safety, and applied machine learning are characterized by publication cycles measured in weeks rather than years, rapid terminological churn, and frequent supersession of earlier findings. A literature review produced by an LLM whose training data ends several months or a year before the time of writing risks presenting an outdated "state of the art," omitting work that has since superseded cited studies, or — in the worst case

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